Re-analysis Summary Report for Peer-evaluation

Paper ID: **Bruner_ExpEco_2017_amYY**

Re-analyst's ID: **y2gle**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
the likelihood … [of decision error] should decrease with the degree of risk aversion (p. 269.)

The corresponding article

Here you can find the original article: https://osf.io/2efb6/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/8xq95/?view_only=f00ab18c5cf94087994e3bb0740ddad1

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
The data were imported into R version 4.1.1 (R Core Team, 2021). Responses from both the probability variation (PV) and reward variation (RV) tasks were used to calculate an average risk aversion score for each participant. Specifically, the sum of risky choices (coded as 1) from the PV and RV tasks were averaged to compute a single risk aversion score per participant. Next, performance on the lottery variation (LV) task was scored, such that the sum of optimal decisions was counted per participant. Then, to assess whether the likelihood of decision error decreases with the degree of an individual's risk aversion, a linear regression model was fit to the data. More specifically, performance in the LV task was regressed on the risk aversion scores. The single-predictor linear model accounted for a significant portion of the variance ($F(1, 104) = 23.86$, $p < .001$, Adjusted $R2 = 0.18$), and the degree of risk aversion was negatively associated with performance on the LV task ($b = -0.63$, $t = -4.89$, $p < .001$).

The analyst's conclusion: The analysis suggests that the likelihood of decision error is associated with greater risk aversion.

The analyst's categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should not use rank-order tests in your analysis.

Here, the analyst reported the following steps and results of the analysis:
The data were imported into R version 4.1.1 (R Core Team, 2021). Responses from both the probability variation (PV) and reward variation (RV) tasks were used to calculate an average risk aversion score for each participant. Specifically, the sum of risky choices (coded as 1) from the PV and RV tasks were averaged to compute a single risk aversion score per participant. Next, performance on the lottery variation (LV) task was scored, such that the sum of optimal decisions was counted per participant. Then, to assess whether the likelihood of decision error decreases with the degree of an individual's risk aversion, a linear regression model was fit to the data. More specifically, performance in the LV task was regressed on the risk aversion scores. The single-predictor linear model accounted for a significant portion of the variance ($F(1, 104) = 23.86$, $p < .001$, Adjusted $R2 = 0.18$), and the degree of risk aversion was negatively associated with performance on the LV task ($b = -0.63$, $t = -4.89$, $p < .001$).

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/ajp65/